

4COSC002W MATHEMATICS FOR COMPUTING

Week 5 Seminar Tasks

Elementary Logic

READING

Lecture 4 Notes (available on Blackboard)

Chapter 15. Croft, T and Davison R (2016) *Foundation maths*, 6th ed. Harlow: Pearson.

Chapter 4. Grossman, P., 2017. *Discrete mathematics for computing*. Bloomsbury Publishing.

TASK 1. LOGIC AND PROPOSITIONS

See Lectures 4 Notes – Slides 4-6

IN-CLASS ACTIVITY

Exercises 15.1 and **15.2** from *Chapter 15. Croft, T and Davison R (2016) Foundation maths*, 6th ed. Harlow: Pearson.

Exercise 15.1



1. Determine which of the following are propositions:

- (a) The moon is made of cheese.
- (b) Swimming is a good exercise.
- (c) Rectangles have four sides.
- (d) Hand in your homework tomorrow.
- (e) Your homework is due in tomorrow.

SENTENCE HAVING
TRUE OR FALSE IS
PROPOSITIONS

YES

YES

YES

NO

YES

Exercise 15.2

1. Given

W : The word has less than seven characters

C : The program compiles

state in words

- (a) $\neg W$ (b) $\neg C$ (c) $W \rightarrow C$ (d) $W \vee \neg C$ (e) $C \wedge W$

2. Given

D : The first character is a digit

V : The word has six characters

state symbolically

- (a) The first character is not a digit.
 (b) If the word has six characters then the first character is a digit.
 (c) The word does not have six characters or the first character is a digit.

1) ATOMIC/ PRIMITIVE (p,q)

2) Compound ($p \wedge q$) / $\sim p$

Exercise 15.2

Question 1

Given:

- W : The word has less than seven characters.
- C : The program compiles.

You are asked to express logical statements in words.

Symbolic form	Meaning in words
(a) $\neg W$	The word does not have less than seven characters. (So it has seven or more characters.) TW
(b) $\neg C$	The program does not compile. TC
(c) $W \rightarrow C$	If the word has less than seven characters, then the program compiles.
(d) $W \vee \neg C$	The word has less than seven characters, or the program does not compile. WVTC
(e) $C \wedge W$	The program compiles and the word has less than seven characters.

Question 2

Given:

- D : The first character is a digit.
- V : The word has six characters.

You are asked to express statements symbolically.

In words	Symbolic form
(a) The first character is not a digit.	$\sim D$ TD
(b) If the word has six characters, then the first character is a digit.	$V \rightarrow D$
(c) The word does not have six characters, or the first character is a digit.	$V \vee \neg D$ TVVD

💡 Note: Statements (b) and (c) are logically equivalent — both express the same idea in different forms.

TASK 2. SYMBOLIC LOGIC

See Lectures 4 Notes – Slides 4-6

For the following formal expressions, establish if their grammar is correct or indicate a problem in their logical structure otherwise.

EXPRESSION TO ANALYSE	CORRECT GRAMMAR? YES/NO	YOUR ANSWERS – WHY “YES” OR “NO”
P	YES	BECAUSE
$\Rightarrow p$		
$\neg \Rightarrow p$		
$p \ q$		
$p \wedge q$		
$p \neg q$		
$p \wedge \neg q$		
$(\neg p \vee q$		
$(\neg p \vee)q$		
$\neg(\neg p \wedge \neg q)$		
$(p \Rightarrow q) \wedge (q \Rightarrow p)$		
$(p \Rightarrow (q \wedge r)) \Rightarrow ((p \Rightarrow q) \wedge (p \Rightarrow r))$		

TASK 3. TRUTH TABLES

See Lectures 4 Notes – Slides 7-13

Task 3.1. Build truth tables for the formulae below. You should follow the algorithm described in the lecture (slide 17).

- a) **IN-CLASS ACTIVITY** $\neg(p \Rightarrow p)$
- b) $\neg(q \vee \neg q)$
- c) **IN-CLASS ACTIVITY** $\neg(p \vee \neg p)$
- d) $p \Rightarrow (p \vee q)$
- e) **IN-CLASS ACTIVITY** $(s \wedge t) \Rightarrow s$
- f) $u \Rightarrow (u \vee w)$

Look at the structure of the given formulae and their truth tables – can you see some patterns here? Can you identify those expressions with similar logical structures and compare their truth tables?

TASK 4. TRUTHS TABLES

See Lectures 4 Notes – Slides 7-13

For each formula below, fill in the relevant information in the table below.

- Based on its truth table, establish if the given formula is satisfiable or not.
- If the given formula is satisfiable, establish if it is valid

If the given formula is satisfiable but not valid, establish at least one combination of truth values for its variables that falsify the formula

IN-CLASS ACTIVITY: 1, 2, 7

	GIVEN FORMULA	FROM THE TRUTH TABLE - IS THE GIVEN FORMULA SATISFIABLE?	VALID/NOT VALID	INPUTS THAT MAKE FORMULA FALSE
1	$\neg (q \vee r) \Rightarrow \neg q \wedge \neg r$			
2	$\neg (q \wedge r) \Rightarrow \neg q \wedge \neg r$			
3	$\neg (q \vee r) \Rightarrow \neg q \vee \neg r$			
4	$\neg (q \wedge r) \Rightarrow \neg q \vee \neg r$			
5	$((q \Rightarrow r) \wedge q) \Rightarrow r$			
6	$((q \Rightarrow r) \wedge r) \Rightarrow q$			
7	$((q \Rightarrow r) \wedge \neg r) \Rightarrow \neg q$			
8	$((q \Rightarrow r) \wedge \neg q) \Rightarrow \neg r$			
9	$p \Rightarrow (q \vee \neg q)$			
10	$(\neg q \wedge q) \Rightarrow \neg r$			
11	$((q \Rightarrow r) \wedge (r \Rightarrow s)) \Rightarrow (q \Rightarrow s)$			

TO BE SATISFIABLE ATLEAST ONE OUTPUT SHOULD BE TRUE
TO BE VALID ALL THE OUTPUTS SHOULD BE TRUE

TASK 5. LOGICAL CONSEQUENCE

See Lectures 4 Notes – Slides 23-30

Check if statement **A** has a logical consequence from knowledge base **KB** in the cases presented in the table below.

You should follow the algorithm for establishing logical consequences, fill the table and consider relevant truth tables for the formulae constructed.

Note: comma (“,”) in KB means conjunction (“and”).

IN-CLASS ACTIVITY: 7, 8, 9

	KB	A	CONSTRUCT LOGICAL CONSEQUENCE FORMULA	FROM THE TRUTH TABLE - IS LOGICAL CONSEQUENCE FORMULA VALID?	IS THERE LOGICAL CONSEQUENCE?
1	p	p			
2	p	$p \vee q$			
3	p, q	p			
4	$\neg (q \vee r)$	$\neg q \wedge \neg r$			
5	$q \Rightarrow r, q$	r			
6	$q \Rightarrow r, r$	q			
7	$q \Rightarrow r, \neg r$	$\neg q$	COMBINE ALL WITH AND OPERATOR AND IMPLIES TO A		
8	$q \Rightarrow r, \neg q$	$\neg r$			
9	p	$q \vee \neg q$			
10	$\neg q \wedge q$	$\neg r$			
11	$q \Rightarrow r, r \Rightarrow s$	$q \Rightarrow s$			

[CHALLENGE] LOGIC GATES WITH MOD 2 ARITHMETIC

In elementary logic, we learned how to use logical operators like AND, OR, IMPLIES, and NOT to combine truth values (True or False) and evaluate logical statements. When we apply these operators to computer science, we often refer to them as logic gates. Logic gates are the building blocks of digital circuits that control how computers process information.

Each logic gate takes inputs (usually True or False, represented as 1 or 0 in binary) and produces an output based on the operation it performs. We can use mod 2 arithmetic (which means calculations where the result is always 0 or 1) to explain how these gates work.

Let's explore how conjunction (AND), disjunction (OR), implication (IMPLIES), and negation (NOT) work using truth tables in mod 2 arithmetic.

Conjunction (AND)

The AND gate outputs 1 only if both inputs are 1. We will break this down using mod 2 arithmetic.

$$A \wedge B = A \times B \bmod 2$$

A	B	$A \times B$	Step-by-Step Calculation	$A \times B \bmod 2$
0	0	0	$0 \times 0 = 0$	$0 \bmod 2 = 0$
0	1	0	$0 \times 1 = 0$	$0 \bmod 2 = 0$
1	0	0	$1 \times 0 = 0$	$0 \bmod 2 = 0$
1	1	1	$1 \times 1 = 1$	$1 \bmod 2 = 1$

Disjunction (OR)

The OR gate outputs 1 if at least one of the inputs is 1. This is equivalent to addition mod 2.

$$A \vee B = (A + B + A \times B) \bmod 2$$

A	B	$A + B + A \times B$	Step-by-Step Calculation	$(A + B + A \times B) \bmod 2$
0	0	0	$0 + 0 + (0 \times 0) = 0$	$0 \bmod 2 = 0$
0	1	1	$0 + 1 + (0 \times 1) = 1$	$1 \bmod 2 = 1$
1	0	1	$1 + 0 + (1 \times 0) = 1$	$1 \bmod 2 = 1$
1	1	3	$1 + 1 + (1 \times 1) = 3$	$3 \bmod 2 = 1$

Implication (IMPLIES)

The IMPLIES gate is true unless A is true and B is false.

$$A \rightarrow B = (1 + A \times (1 - B)) \bmod 2$$

A	B	$1 + A \times (1 - B)$	Step-by-Step Calculation	$(1 + A \times (1 - B)) \bmod 2$
0	0	1	$1 + (0 \times (1 - 0)) = 1$	$1 \bmod 2 = 1$
0	1	1	$1 + (0 \times (1 - 1)) = 1$	$1 \bmod 2 = 1$
1	0	2	$1 + (1 \times (1 - 0)) = 2$	$2 \bmod 2 = 0$
1	1	1	$1 + (1 \times (1 - 1)) = 1$	$1 \bmod 2 = 1$

Negation (NOT)

The NOT gate flips the input value. In mod 2 arithmetic, this is done by subtracting A from 1.

$$\neg A = (1 - A) \bmod 2$$

A	$1 - A$	Step-by-Step Calculation	$(1 - A) \bmod 2$
0	1	$1 - 0 = 1$	$1 \bmod 2 = 1$
1	0	$1 - 1 = 0$	$0 \bmod 2 = 0$

TASK: MOD 2 LOGIC GATE CALCULATION

For the following logical expressions, calculate the results using mod 2 arithmetic:

- $(A \wedge B) \rightarrow (A \vee B)$
- $A \vee (\neg B) \wedge A$